Frederick Joseph Boehm, III

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Academic Appointment

South Dakota State University — Brookings, South Dakota, USA
Assistant Professor, Department of Mathematics and Statistics — 08/2024 to present

Education and Training

University of Michigan — Ann Arbor, Michigan, USA
Postdoctoral Research Fellow, Department of Internal Medicine (NIH T32 HL007853) — 07/2023 to 06/2024
Postdoctoral Research Fellow, Department of Biostatistics — 06/2021 to 05/2023

University of Massachusetts Medical School — Worcester, Massachusetts, USA
Postdoctoral Research Fellow, Department of Microbiology & Physiological Systems — 09/2019 to 05/2021

University of Wisconsin-Madison — Madison, Wisconsin, USA
Postdoctoral Research Fellow, Department of Biostatistics and Medical Informatics — 06/2019 to 08/2019
Ph.D., Department of Statistics — 09/2011 to 05/2019
Postdoctoral Research Fellow (NIH T32 ES007015), Molecular and Environmental Toxicology Center 02/2012 to 01/2014
Assistant Researcher, Value-Added Research Center — 01/2011 to 07/2011

University of Washington — Seattle, Washington, USA
Postdoctoral Research Fellow, Department of Biostatistics — 11/2007 to 09/2009

University of Wisconsin-Madison — Madison, Wisconsin, USA
M.D., School of Medicine and Public Health — 08/2003 to 08/2007
M.S., Department of Population Health Sciences — 01/2003 to 08/2007
Graduate Project Assistant, Department of Population Health Sciences — 01/2003 to 05/2003

La Clinica de los Campesinos — Wautoma, Wisconsin, USA
Spanish Language Medical Interpreter — 06/2002 to 08/2002

University of Wisconsin-Madison — Madison, Wisconsin, USA
B.S., with Honors in Chemistry, Mathematics, and Biochemistry — 09/1996 to 05/2001

Written Works

Unpublished Manuscripts

1. Adebowale Awosika-Olumo, **Frederick J. Boehm**, Rinoj Gautam, and Usha Sambamoorthi (in preparation). Predicting Human Papillomavirus Vaccination Uptake Among U.S. adults Using BRFSS 2023: A Cross-Sectional Machine Learning Study.
2. Alexander Katz, Cameron B. Pinnock, Ayse Bilge Ozel, **Frederick J. Boehm**, Andrea Murad, Min-Lee Yang, Kristina L. Hunker, Natalia Fendrikova Mahlay, Dawn M. Coleman, Matthew Zawistowski, Jeffrey Olin, James Froehlich, Jason Kovacic, Heather Gornik, James C. Stanley, Xiang Zhou, and Santhi K. Ganesh (submitted).

Rare coding variation in cell-matrix adhesion and trimer collagen genes influences disease susceptibility and severity in fibromuscular dysplasia.
3. Peng Zhou, Ji Hoon Park, and **Frederick J. Boehm** (in preparation). ShinyMR: A web platform to promote rigor and reproducibility in Mendelian randomization studies.
4. Emma G. Glackin and **Frederick J. Boehm** (in preparation). Preferential sampling of environmental monitoring networks method comparison: A Bayesian MCMC framework and INLA.
5. Christine Larson and **Frederick J. Boehm** (in preparation). Visible Thinking in Calculus.
6. Jose L. Gonzalez Hernandez, **Frederick J. Boehm**, Radhey Kaushik, Phibi Naduvathu, Jeff M. Martin, Philip M. Urso, and Michael Gonda (in preparation). Genomic Erosion and Structured Inbreeding Within a Founding North American Bison Herd.
7. Kamyar Sharifi, Usha Sambamoorthi, Grace Williams, eMERGE Network, **Frederick J. Boehm**, and David R. Crosslin (in preparation). Genetic and Clinical Differences in Asthma First Recorded in Childhood and Adulthood.

Journal Articles

8. Christina G. Hutten, **Frederick J. Boehm**, Jennifer A. Smith, Brian W. Spitzer, Sylvia Wassertheil-Smoller, Carmen R. Isasi, Jianwen Cai, Jonathan T. Unkart, Jiehuan Sun, Victoria Persky, Martha L. Daviglus, Tamar Sofer, and Maria Argos (2025). Differential performance of polygenic risk scores for heart disease in Hispanic/Latino subgroups: Findings of the Hispanic Community Health Study/Study of Latinos. *Human Genetics and Genomics Advances* 6(4):100486. https://doi.org/10.1016/j.xhgg.2025.100486.
9. Sherry L. Kurtz, Richard E. Baker, **Frederick J. Boehm**, Chelsea C. Lehman, Lara R. Mittereder, Hamda Khan, Amy P. Rossi, Daniel M. Gatti, Gillian Beamer, Christopher M. Sassetti, and Karen L. Elkins (2024). Multiple genetic loci influence vaccine-induced protection against Mycobacterium tuberculosis in genetically diverse mice. *PLOS Pathogens* 20(3):e1012069. https://doi.org/10.1371/journal.ppat.1012069.
10. **Frederick J Boehm** and Xiang Zhou (2022). Statistical methods for Mendelian randomization in genome-wide association studies: A review. *Computational and Structural Biotechnology Journal* 20:2338-2351. https://doi.org/10.1016/j.csbj.2022.05.015.
11. Clare M Smith, Richard E Baker, Megan K Proulx, Bibhuti B Mishra, Jarukit E Long, Sae Woong Park, Ha-Na Lee, Michael C Kiritsy, Michelle M Bellerose, Andrew J Olive, Kenan C Murphy, Kadamba Papavinasasundaram, **Frederick J Boehm**, Charlotte J Reames, Rachel K Meade, Brea K Hampton, Colton L Linnertz, Ginger D Shaw, Pablo Hock, Timothy A Bell, Sabine Ehrt, Dirk Schnappinger, Fernando Pardo-Manuel de Villena, Martin T Ferris, Thomas R Ioerger, and Christopher M Sassetti (2022). Host-pathogen genetic interactions underlie tuberculosis susceptibility in genetically diverse mice. *Elife* 11:e74419. https://doi.org/10.7554/eLife.74419.
12. **Frederick J Boehm** and Bret M Hanlon (2021). What is happening on Twitter? A framework for student research projects with tweets. *Journal of Statistics and Data Science Education* 29:S95-S102. https://doi.org/10.1080/10691898.2020.1848486.
13. Jorge L Rodriguez-Gil, Dawn E Watkins-Chow, Laura L Baxter, Gene Elliot, Ursula L Harper, Stephen M Wincovitch, Julia C Wedel, Arturo A Incao, Mylene Huebecker, **Frederick J Boehm**, William S Garver, Forbes D Porter, Karl W Broman, Frances M Platt, and William J Pavan (2020). Genetic background modifies phenotypic severity and longevity in a mouse model of Niemann-Pick disease type C1. *Disease Models & Mechanisms* 13(3):dmm.042614. https://doi.org/10.1242/dmm.042614.
14. **Frederick J. Boehm**, Elissa J. Chesler, Brian S. Yandell, and Karl W. Broman (2019). Testing pleiotropy vs. separate QTL in multiparental populations. *G3* 9(7):2317. https://doi.org/10.1534/g3.119.400098.
15. Julia H Kemis, Vanessa Linke, Kelsey L Barrett, **Frederick J Boehm**, Lindsay L Traeger, Mark P Keller, Mary E Rabaglia, Kathryn L Schueler, Donald S Stapleton, Daniel M Gatti, Gary A Churchill, Daniel Amador-Noguez, Jason D Russell, Brian S Yandell, Karl W Broman, Joshua J Coon, Alan D Attie, and Federico E Rey (2019).

Genetic determinants of gut microbiota composition and bile acid profiles in mice. *PLoS genetics* 15(8):e1008073. https://doi.org/10.1371/journal.pgen.1008073.

16. **Frederick Boehm**, Brian Yandell, and Karl W Broman (2019). qtl2pleio: Testing pleiotropy vs. separate QTL in multiparental populations. *Journal of Open Source Software* 4(38):1435. https://doi.org/10.21105/joss.01435.

17. Chris Wells, Dhavan V. Shah, Jon C. Pevehouse, JungHwan Yang, Ayellet Pelled, **Frederick Boehm**, Josephine Lukito, Shreenita Ghosh, and Jessica L. Schmidt (2016). How Trump Drove Coverage to the Nomination: Hybrid Media Campaigning. *Political Communication* 33(4):669-676. https://doi.org/10.1080/10584609.2016.1224416.

18. Eric C Tauchman, **Frederick J Boehm**, and Jennifer G DeLuca (2015). Stable kinetochore-microtubule attachment is sufficient to silence the spindle assembly checkpoint in human cells. *Nature communications* 6(1):10036. https://doi.org/10.1038/ncomms10036.

19. William F Dove, Alexandra Shedlovsky, Linda Clipson, James M Amos-Landgraf, Richard B Halberg, Kathleen J Krentz, **Frederick J Boehm**, Michael A Newton, David J Adams, and Thomas M Keane (2014). A strategy to identify dominant point mutant modifiers of a quantitative trait. *G3: Genes| Genomes| Genetics* 4(6):1113–1121. https://doi.org/10.1534/g3.114.010595.

20. James C Costello, Laura M Heiser, Elisabeth Georgii, Mehmet Gönen, Michael P Menden, Nicholas J Wang, Mukesh Bansal, Petteri Hintsanen, Suleiman A Khan, John-Patrick Mpindi, et al. (2014). A community effort to assess and improve drug sensitivity prediction algorithms. *Nature biotechnology* 32(12):1202–1212. https://doi.org/10.1038/nbt.2877.

21. Cathy C Laurie, Kimberly F Doheny, Daniel B Mirel, Elizabeth W Pugh, Laura J Bierut, Tushar Bhangale, **Frederick Boehm**, Neil E Caporaso, Marilyn C Cornelis, Howard J Edenberg, Stacy B Gabriel, Emily L Harris, Frank B Hu, Kevin B Jacobs, Peter Kraft, Maria Teresa Landi, Thomas Lumley, Teri A Manolio, Caitlin McHugh, Ian Painter, Justin Paschall, John P Rice, Kenneth M Rice, Xiuwen Zheng, Bruce S Weir, and the GENEVA Investigators (2010). Quality control and quality assurance in genotypic data for genome-wide association studies. *Genetic epidemiology* 34(6):591–602. https://doi.org/10.1002/gepi.20516.

22. MC Smith Fawzi, W Lambert, **F Boehm**, JL Finkelstein, JM Singler, F Léandre, P Nevil, D Bertrand, MS Claude, J Bertrand, M Louissaint, L Jeannis, PE Farmer, AT Yang, and JS Mukherjee (2010). Economic Risk Factors for HIV Infection Among Women in Rural Haiti: Implications for HIV Prevention Policies and Programs in Resource-Poor Settings. *Journal of Women's Health* 19(5):885–892. https://doi.org/10.1089/jwh.2008.1334.

23. Elizabeth E Marchani, Yanming Di, Yoonha Choi, Charles Cheung, Ming Su, **Frederick Boehm**, Elizabeth A Thompson, and Ellen M Wijsman (2009). Contrasting identity-by-descent estimators, association studies, and linkage analyses using the Framingham Heart Study data. *BMC proceedings* 3(Suppl 7):S102. https://doi.org/10.1186/1753-6561-3-S7-S102.

24. Jason E Gestwicki, Laura E Strong, Christopher W Cairo, **Frederick J Boehm**, and Laura L Kiessling (2002). Cell aggregation by scaffolded receptor clusters. *Chemistry & biology* 9(2):163–169. https://doi.org/10.1016/S1074-5521(02)00102-3.

Edited Book

25. A Vishnu Moorthy, Bryan Neil Becker, **Frederick J Boehm III**, and Arjang Djamali, editors (2009). *Pathophysiology of Kidney Disease and Hypertension*. Saunders/Elsevier.

Book Chapters

26. Ayellet Pelled, Josephine Lukito, **Frederick Boehm**, JungHwan Yang, and Dhavan Shah (2018). `Little Marco',` Lyin' Ted', `Crooked Hillary', and the` Biased' Media: How Trump Used Twitter to Attack and Organize. In Natalie Stroud and Shannon McGregor, editors *Digital Discussions: How Big Data Informs Political Communication*. Routledge.

27. Vishnu Moorthy and **Frederick Boehm** (2009). Urinalysis and an Approach to Kidney Diseases. In Vishnu Moorthy, Bryan Becker, **Frederick Boehm**, and Arjang Djamali, editors *Pathophysiology of Kidney Disease and Hypertension*. Saunders/Elsevier.
28. Kelly Traeger, **Frederick Boehm**, and Arjang Djamali (2009). Acid-Base Homeostasis and Metabolic Alkalosis. In Vishnu Moorthy, Bryan Becker, **Frederick Boehm**, and Arjang Djamali, editors *Pathophysiology of Kidney Disease and Hypertension*. Saunders/Elsevier.
29. **Frederick Boehm**, Kelly Traeger, and Arjang Djamali (2009). Metabolic Acidosis and Approach to Acid-Base Disorders. In Vishnu Moorthy, Bryan Becker, **Frederick Boehm**, and Arjang Djamali, editors *Pathophysiology of Kidney Disease and Hypertension*. Saunders/Elsevier.

Software

30. **Frederick Boehm** (2018-06-04). qtl2pleio R package. https://github.com/fboehm/qtl2pleio.
31. **Frederick Boehm** (2018-03-15). gemma2 R package. https://github.com/fboehm/gemma2.

Conference Proceedings

32. J Mark, L Wesler, C Oswald, F Léandre, P Nevil, D Bertrand, J Bertrand, **F Boehm**, and MC Smith Fawzi (2011). P2-S9. 07 Economic risk factors for syphilis infection among pregnant women in rural Haiti. In *Sexually Transmitted Infections*.
33. JE Gestwicki, CW Cairo, **FJ Boehm**, and LL Kiessling (2001). Mechanistic origins of the valency effect: Multivalent ligand binding studied by surface plasmon resonance and isothermal titration calorimetry.. In *ABSTRACTS OF PAPERS OF THE AMERICAN CHEMICAL SOCIETY*.
34. E Fasella, **FJ Boehm**, and LL Kiessling (2000). Synthesis of galactosyl-alpha (1-3)-galactosyl epitopes for use in immunological studies.. In *ABSTRACTS OF PAPERS OF THE AMERICAN CHEMICAL SOCIETY*.

Ph.D. Thesis

35. **Frederick J. Boehm III** (2019). Testing pleiotropy vs. separate QTL in multiparental populations.

Other Publications

36. Catherine DeRose, Carrie Roy, and **Fred Boehm** (2014). Victorian eyes: Literary, statistical, and artistic perspectives on Victorian novels – and Dickens's unfinished murder mystery. *Significance*.
37. **Fred Boehm** (1997). The lone male wolf of the Sandhill Wildlife Area. *International Wolf*.

Research Funding

Funded

Conference: SDSU Data Science Symposium December 2024
Principal investigator · National Science Foundation · Total direct costs: $7,502 · Award No. 2515675

Under Review

Targeting POLR1A-dependent ribosome biogenesis as a new therapeutic approach for advanced liver cancer June 2026
Co-investigator (PI: Bilal Hafeez) · National Institutes of Health (R15) · Total direct costs: $375,000
Bayesian Hierarchical Modeling for Multi-trait, Multi-ancestry Polygenic Scores June 2026
Principal investigator · National Institutes of Health (R15) · Total direct costs: $361,945

Not Funded

Measuring the accuracy of multi-trait, multi-ancestry polygenic scores March 2026
Principal investigator · South Dakota Board of Regents (Competitive Research Grant) · Total direct costs: $86,959
Collaborative Research: FEC: StaTIC: Structure- and Data-Driven Discovery of Molecular Glues for Transient Transcription-Factor Complexes January 2026
Co-investigator (PI: Suvobrata Chakravarty) · National Science Foundation (EPSCoR FEC) · Total direct costs: $3,130,401
Expanding and Analyzing Genetic Diversity Across Bison Herds November 2025
Co-investigator (PI: Jose Gonzalez) · USDA NIFA · Total direct costs: $454,970
Measuring the impact of linkage disequilibrium panels on polygenic score accuracy November 2025
Principal investigator · Healthy Americas Research Consortium · Total direct costs: $10,000
Measuring the Impact of LD Reference Panel Choice on PGS Accuracy in Admixed Populations November 2025
Principal investigator · BioSNTRii · Total direct costs: $75,000
Ultra-processed Food Exposure in the USA: Advancing Biological and Behavioral Evidence to Inform Dietary Practice November 2025
Co-investigator (PD: Moul Dey; Co-PD: J. Roberts) · USDA NIFA · Total costs: $649,215

Teaching Experience

Formal Courses

Instructor, Bioinformatics, South Dakota State University Fall 2026
Guest lecturer, Career Seminar, South Dakota State University Spring 2025
Instructor, Senior Capstone I, South Dakota State University Spring 2025
Instructor, Statistical methods II, South Dakota State University Spring 2025, Spring 2026
Instructor, Time series analysis, South Dakota State University Fall 2024, Fall 2025, Fall 2026
Co-facilitator, Inclusive teaching for educating a diverse nation, University of Wisconsin-Madison Delta Program & CIRTL Network Spring 2017, Fall 2017
Guest lecturer on topic models & Teaching assistant, Special Topics: Data Science, University of Wisconsin-Madison Fall 2016
Instructor, Introductory statistics, University of Wisconsin-Madison Fall 2016, Spring 2016, Fall 2015, Summer 2015, Spring 2015, Fall 2014, Spring 2014
Teaching assistant, Introduction to Biostatistics, University of Washington Fall 2009, Winter 2010
Teaching assistant, Introductory organic chemistry, University of California, Berkeley Fall 2001

Short Courses & Workshops

Instructor, Engaging students in the teaching of statistics, CIRTL Network March 2018, July 2020
Instructor, Data Carpentry Workshop, University of Wisconsin-Madison, Missouri S&T, George Mason University June 2017, September 2018, January 2019
Instructor, Data analysis with topic models for communications researchers, University of Wisconsin-Madison July 2016, October 2016
Instructor, Topic modeling for non-statisticians, University of Wisconsin-Madison October 2013
Instructor, Statistical computing with R, University of Wisconsin-Madison June 2012, June 2013

Training in Teaching

Postdoctoral Short Course on College Teaching in STEM, University of Michigan Center for Research on Learning and Teaching Spring 2024
Expeditions in Learning: Exploring How Students Learn with Writing Across the Curriculum, University of Wisconsin-Madison Delta Program Spring 2016
Diversity in the College Classroom, University of Wisconsin-Madison Delta Program Fall 2015
Research Mentor Training, University of Wisconsin-Madison Delta Program Summer 2016 & Summer 2012
An Introduction to Evidence-Based Undergraduate STEM Teaching, CIRTL Network Fall 2014
The College Classroom, University of Wisconsin-Madison Delta Program Summer 2014
Instructional Materials Development: Integrating Statistics into STEM courses, University of Wisconsin-Madison Delta Program Spring 2014
Teaching Statistics in the Classroom, University of Wisconsin-Madison Delta Program Spring 2013

Mentoring Experience

Research mentor to Robert Turner, undergraduate researcher May 2015 to May 2016
Project: Topic Modeling of Tweets to Detect Social Media Events
Next position: Software Engineer at Epic Systems
Research mentor to Jinyu Xia, undergraduate researcher September 2015 to August 2016
Project: Tweet Sentiments and Stock Market Index Trends
Next position: Statistics graduate student at University of Washington
Research mentor to Ji Hoon Park, undergraduate researcher November 2024 to May 2025
Project: Building a Shiny App for Mendelian Randomization Analysis
Next position: Graduate student researcher
Research mentor to Emma Glackin, graduate researcher June 2025 to present
Project: Causal Inference Methods in Environmental Epidemiology
Research mentor to Etornam Kunu, graduate researcher September 2025 to present
Project: Polygenic Scores in Diverse Populations
Research mentor to Shree Nyaupane, graduate researcher September 2025 to present
Project: Rare Variants in Complex Traits and Diseases
Research mentor to Henrietta Kotey, graduate researcher September 2025 to May 2026
Project: Multivariable Mendelian Randomization Methods
Research mentor to Peng Zhou, Gradaute researcher June 2026 to present
Project: ShinyMR: A web application to promote rigor and reproducibility in Mendelian randomization studies

Leadership Training

Participant, University of Michigan You3 Leadership Training Program Spring 2024
Participant, University of Michigan Designing Your Life Course Fall 2022
Participant, Leadership and Management in Action Program Fall 2021
Committee Member, Genetics Society of America Early Career Researchers Leadership Program February 2020 to February 2022
Participant, American Statistical Association Workshop on Communicating with the Media and the Public November 2016

Awards & Honors

Conference Travel Award, University of Michigan Postdoc Association 2024
Best Poster Award, International Mammalian Genome Conference 2018
Conference Travel Scholarship, International Mammalian Genome Conference 2018
Integrating Research Ethics and Scholarship Fellowship, University of Wisconsin-Madison 2013
New Arts Venture Challenge Award (for Victorian Eyes art exhibition), University of Wisconsin-Madison 2013
Medical Student Research Training Fellowship, Howard Hughes Medical Institute 2006
Medical Student Summer Research Stipend, University of Wisconsin-Madison Department of Family Medicine 2004
Graduate Research Fellowship, National Science Foundation 2001
Meyerhoff Undergraduate Excellence Award, University of Wisconsin-Madison 2001
University Book Store Award for Outstanding Undergraduate Thesis, University of Wisconsin-Madison 2001
Edward Panek Memorial Scholarship, University of Wisconsin-Madison Department of Chemistry 2000
Elected as Member, The Phi Beta Kappa Society 1999
Wisconsin Idea Fellowship (for the Salvation Army Learning Center), University of Wisconsin-Madison 1999
Dean's List, University of Wisconsin-Madison 1998 to 2001
Summer Research Scholarship, University of Wisconsin-Madison Department of Chemistry 1998
College of Letters & Science Honors Program Member, University of Wisconsin-Madison 1996 to 2001
Wisconsin Academic Excellence Scholarship, University of Wisconsin-Madison 1996 to 2000

Presentations

Slide Presentations

Does the LD panel matter for PGS? Ancestry and size of LD reference panels for polygenic scoring May 28, 2026
American Statistical Association Midwest Regional Conference in Statistics and Data Science (Invited)
Minneapolis, Minnesota, United States of America

Exploring Human Biology with Genome-Wide Association Studies December 03, 2025
Sanford Research Seminar Series (Invited)
Sioux Falls, South Dakota, United States of America

Exploring Human Biology with Genome-Wide Association Studies October 31, 2025
South Dakota State University Corothers Lecture (Invited)
Brookings, South Dakota, United States of America

My Research Goals September 15, 2025
South Dakota State University New Faculty Forum
Brookings, South Dakota, United States of America

Introduction to our Research Team August 19, 2025
South Dakota State University Graduate Student Orientation
Brookings, South Dakota, United States of America

shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies August 04, 2025
Joint Statistical Meetings 2025
Nashville, Tennessee, United States of America

shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies July 31, 2025
Institute for Mathematical Statistics New Researchers Conference 2025
Nashville, Tennessee, United States of America

shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies May 21, 2025
STATGEN2025 Conference

Minneapolis, Minnesota, United States of America

A Shiny App to Promote Rigor and Reproducibility in Mendelian Randomization Studies February 07, 2025
South Dakota State University Data Science Symposium (Invited)
Brookings, South Dakota, United States of America

Distinguishing pleiotropy from separate QTL in multiparental populations July 30, 2018
Joint Statistical Meetings
Vancouver, British Columbia, Canada

Distinguishing pleiotropy from separate QTL in multiparental populations June 20, 2018
Complex Trait Community Meeting
Glasgow, Scotland, United Kingdom

Event Detection with Social Media Data June 30, 2016
UseR! 2016 Meeting
Stanford, California, United States of America

Post-GWAS characterization of breast cancer-associated SNPs November 28, 2012
University of Wisconsin-Madison Department of Statistics Student Seminar
Madison, Wisconsin, United States of America

Identification of Expression Quantitative Trait Loci among breast cancer-associated SNPs September 20, 2012
University of Wisconsin-Madison Molecular and Environmental Toxicology Center Seminar
Madison, Wisconsin, United States of America

Poster Presentations

shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies July 31, 2025
Institute for Mathematical Statistics New Researchers Conference 2025
Nashville, Tennessee, United States of America

shinyMR: An App to Promote Rigor and Reproducibility in Mendelian Randomization Studies July 07, 2025
Human Genetics and Genomics Gordon Conference
Portland, Maine, United States of America

Testing pleiotropy vs. separate QTL in multiparental populations November 11, 2018
International Mammalian Genome Conference
Rio Mar, Puerto Rico, United States of America

Testing pleiotropy vs. separate QTL in multiparental populations October 26, 2018
University of Wisconsin-Madison Computation and Informatics in Biology and Medicine Training Program Retreat
Madison, Wisconsin, United States of America

Event Detection with Topic Modeling of Tweets July 13, 2017
Society for Political Methodology's Annual Summer Conference (PolMeth XXXIV)
Madison, Wisconsin, United States of America

Pleiotropy vs. Close Linkage in the Diversity Outbred Mouse Population October 22, 2016
The Jackson Laboratory Systems Genetics Short Course
Bar Harbor, Maine, United States of America

Pleiotropy vs. Close Linkage in the Diversity Outbred Mouse Population October 07, 2016
University of Wisconsin-Madison Computation and Informatics in Biology and Medicine Training Program Retreat
Madison, Wisconsin, United States of America

Data Analysis with Topic Models for Communications Researchers August 06, 2016
Association for Education in Journalism and Mass Communication Annual Meeting

Minneapolis, Minnesota, United States of America

Event Detection with Topic Modeling of Tweets April 06, 2016
University of Wisconsin-Madison WARF Discovery Challenge Symposium
Madison, Wisconsin, United States of America

Identification of Expression Quantitative Trait Loci among breast cancer-associated SNPs October 12, 2012
University of Wisconsin-Madison Computation and Informatics in Biology and Medicine Retreat
Madison, Wisconsin, United States of America

Identification of Expression Quantitative Trait Loci among breast cancer-associated SNPs May 22, 2012
University of Wisconsin-Madison Molecular and Environmental Toxicology Center Retreat
Madison, Wisconsin, United States of America

Service

University Service

Thesis Committee Member for Wilberforce Arthur, South Dakota State University May 2026
South Dakota State University College of Engineering Day Participant, South Dakota State University College of Engineering September 2024, October 2025
Outreach & Recruitment Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2025 to present
Seminar Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2025 to present
Co-organizer, South Dakota State University Data Science Symposium February 2025
Statistics Curriculum Revision Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2024 to present
Data Science Curriculum Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2024 to May 2025
Senior Capstone Committee Member, Department of Mathematics and Statistics, South Dakota State University September 2024 to May 2025
Co-coordinator, University of Wisconsin-Madison Computational Methods in Mass Communications Research Group September 2016 to December 2016
Coordinator, University of Wisconsin-Madison WISCIENCE Graduate Student Book Group June 2015 to May 2016
Organizer, University of Wisconsin-Madison Data, Humanities, and Art (DH + A) Symposium September 2013 to November 2013
Moderator, University of Wisconsin-Madison Undergraduate Research Symposium April 2013, April 2012
Member, University of Wisconsin-Madison Graduate Student Peer Mentor Award Review Committee April 2013
Member, University of Wisconsin-Madison Graduate School Vilas Travel Grant Review Committee October 2012 to November 2012
Member, University of Wisconsin-Madison Statistics Department Social Committee January 2012 to May 2014

Community Service

Coach, Westborough Mathcounts Team September 2020 to April 2022
Penpal, Letters to a Pre-Scientist September 2018 to May 2024
Coach, Madison Science Olympiad Team November 2013 to March 2014
Assistant coach, Madison Mathcounts Team September 2012 to May 2013

Volunteer, MEDiC	September 2003 to May 2005
Founding director, Salvation Army Learning Center	September 1998 to May 2000
Cub Scouts Den Leader, Boy Scouts of America	September 1998 to June 2000

Professional Service

Peer reviewer for abstracts submitted to the ASHG Annual Meeting, American Society for Human Genetics June 2026

Peer reviewer for Research Experiences for Undergraduates (REU) grant proposals, National Science Foundation May 2026

Co-Secretary, American Statistical Association JEDI Outreach Group	January 2026 to present
Session Chair, Joint Statistical Meetings	August 2025
Session Chair, ENAR Annual Meeting	March 2025
Ad hoc Reviewer, National Science Foundation	December 2024
DNA Day Essay Judge, American Society for Human Genetics	March 2024
Participant, ENAR Fostering Diversity in Biostatistics Virtual Workshop	March 2021
Associate Editor, Journal of Open Source Software	June 2020 to present

Peer Reviewer, Journal of Open Source Software, European Journal of Epidemiology, The R Journal, ROpenSci, Genetics, G3, PLoS Computational Biology, BMC Cancer, Journal of Statistics and Data Science Education January 2019 to present

Trainee, Genetics Society of America Peer Review Training Program	March 2018 to March 2020
Instructor, The Carpentries	May 2017 to present
Participant, Software Carpentry Instructor Training	October 2016